DATA 581

Modeling and Simulation II

Modelling More than One Random Variable

- **Joint, Marginal and Conditional pdfs**
- **Simple Regression Model**
- **Independence**
- **Covariance, Correlation and Dependence**

Until now, we have been studying properties of a single variable: *univariate* statistics.

Any mention of more than 2 variables at a time has usually been under the assumption that the variables follow the same distribution and are independent.

Much of what makes statistics useful is the study of the relationships between multiple variables.

We start with 2 measurements and consider larger samples subsequently.

Modelling More than One Random Variable

Suppose x_1 and x_2 are 2 observations.

The probability density function (joint pdf) should be a function of 2 variables:

$$f(x_1, x_2).$$

In order for valid probabilities to be computed, it is necessary that this function be nonnegative, and its total (double) integral be 1.

Probability Calculations

Probabilities are calculated by integration.

The probability that X_1 lies in the interval (a, b) and X_2 lies in the interval (c, d) is

$$P(X_1 \in (a, b), X_2 \in (c, d)) = \int_a^b \int_c^d f(x_1, x_2) dx_1 dx_2.$$

Example

Suppose that there are 2 measurements that are uniformly distributed and the joint probability density function for these measurements is

$$f(x_1, x_2) = \frac{1}{(b-a)(d-c)}$$

when $x_1 \in [a, b]$ **and** $x_2 \in [c, d]$**, and 0, otherwise. Using calculus, it can be shown that**

$$\int_c^d \int_a^b f(x_1, x_2) dx_1, dx_2 = 1.$$

Suppose $c = a = 0$ **and** $d = b = 1$**. The probability that both measurements exceed 0.75 is**

$$\int_{0.75}^1 \int_{0.75}^1 dx_1 dx_2 = (1 - 0.75)^2 = .0625.$$

Another Example

Consider the function

$$f(x_1, x_2) = \frac{\lambda}{x_1} e^{-\lambda x_1 - x_2/x_1}, \qquad x_1, x_2 \geq 0$$

and 0, otherwise.

$f(x_1, x_2) \geq 0$ everywhere, and

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2) dx_1 dx_2 = 1.$$

Therefore $f(x_1, x_2)$ is a joint pdf.

Marginal Distributions

Earlier, we saw that

$$P(X_1 \in (a,b), X_2 \in (c,d)) = \int_a^b \int_c^d f(x_1,x_2)dx_1 dx_2.$$

What if we are only interested in $P(X_1 \in (a,b))$?

In this case, it doesn't make a difference to us what the value of X_2 is; it could be any element of $(-\infty, \infty)$.

In other words, we could write

$$P(X_1 \in (a,b)) = P(X_1 \in (a,b), X_2 \in (-\infty,\infty))$$

but, in integral form, this becomes

$$P(X_1 \in (a,b)) = \int_a^b \int_{-\infty}^{\infty} f(x_1,x_2)dx_2 dx_1.$$

Marginal pdf

Recalling how we defined the probability density function for a single random variable, i.e.

$$P(X_1 \in (a,b)) = \int_a^b f(x)dx,$$

it necessarily follows that the probability density function for X_1 must be

$$f(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2)dx_2.$$

Similar reasoning tells us that the probability density function for X_2 must be

$$f(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2)dx_1.$$

To distinguish these probability density functions, if necessary, we write $f_{X_1}(x)$ as the probability density function of X_1 and $f_{X_2}(x)$ as the probability density function of X_2.

$f_{X_1}(x)$ and $f_{X_2}(x)$ are referred to as *marginal* density functions.

Marginal pdf

To summarize:

When in the context of several random variables, the marginal density of a single one of the random variables can be obtained by integrating the joint density function over all possible values of all of the random variables apart from the one of interest.

Example: Dependent Exponential Random Variables

When the joint density function is

$$f(x_1, x_2) = \frac{\lambda}{x_1} e^{-\lambda x_1 - x_2/x_1}, \qquad x_1, x_2 \geq 0$$

we can obtain the marginal density function for X_1 by integrating over all possible values of x_2 (i.e. $x_2 > 0$):

$$f_{X_1}(x_1) = \int_0^\infty \frac{\lambda}{x_1} e^{-\lambda x_1 - x_2/x_1} dx_2 = \lambda e^{-\lambda x_1}, \quad x_1 \geq 0$$

and 0, otherwise. We recognize this as the exponential density function.

Example: Dependent Exponential Random Variables

Attempting to obtain the marginal density function for X_2 results in

$$f_{X_2}(x_2) = \int_0^\infty \frac{\lambda}{x_1} e^{-\lambda x_1 - x_2/x_1} dx_1$$

which can only be evaluated numerically. Monte Carlo integration, anyone?

Clearly, X_2 does not have the same density function as X_1, so they are not identically distributed.

Conditional Density Functions

Suppose we know the value of X_1, and we would like to predict the value of X_2, using this information.

The joint probability density function tells us how X_1 and X_2 are related, so we might think that $f(x_1, x_2)$ is the probability density function of X_2 for each given value of X_1, but this would be an incorrect interpretation.

This is because

$$\int_{-\infty}^{\infty} f(x_1, x_2) dx_2 = f_{X_1}(x_1)$$

If $f(x_1, x_2)$ were a probability density function for X_2, given X_1, the above integral should evaluate to 1.

Conditional Density Functions

However, if we divide $f(x_1, x_2)$ by $f_{X_1}(x_1)$, we obtain a function of x_2 which integrates to 1:

$$\int_{-\infty}^{\infty} \frac{f(x_1, x_2)}{f_{X_1}(x_1)} dx_2 = \frac{f_{X_1}(x_1)}{f_{X_1}(x_1)} = 1.$$

It turns out that this gives a very useful predictive density function for X_2, given knowledge of X_1.

We write

$$f_{X_2|X_1}(x_1, x_2) = \frac{f(x_1, x_2)}{f_{X_1}(x_1)}$$

as the *conditional density function* of X_2 given X_1. This density function predicts the probability density of X_2, when we know the value of X_1.

Conditional Density Functions

Similar reasoning tells us that the predictive probability density of X_1 or its conditional density function is given by

$$f_{X_1|X_2}(x_1, x_2) = \frac{f(x_1, x_2)}{f_{X_2}(x_2)}$$

Example - Conditional Density Functions

Suppose

$$f(x_1, x_2) = \frac{\lambda}{x_1} e^{-\lambda x_1 - x_2/x_1}, \qquad x_1, x_2 \geq 0.$$

We have

$$f_{X_1}(x_1) = \lambda e^{-\lambda x_1}, \quad x_1 \geq 0.$$

Then the conditional density for X_2, given X_1

$$f_{X_2|X_1}(x_1, x_2) = \frac{1}{x_1} e^{-x_2/x_1}, \quad x_2 \geq 0.$$

This is an exponential density function, but now the rate is $1/x_1$. This density function is not the same as the marginal density of X_2. Thus, X_1 gives predictive information about X_2. The two random variables are not independent.

Simple Regression

Suppose X and Y are random variables having a joint density function given by

$$f(x,y) = \frac{\mathrm{e}^{-(y-\beta_0-\beta_1 x)^2/(2\sigma^2)-x^2/2}}{2\pi\sigma}.$$

If we want to predict Y from X, we should use the conditional density function $f_{Y|X}(x,y)$. We can obtain that density function in 2 steps:

1. **Find $f_X(x)$ by integrating over all y.**

2. **Divide $f(x,y)/f_X(x)$.**

Simple Regression

1.

$$f_X(x) = \int f(x,y)dy = \frac{\mathbf{e}^{-x^2/2}}{\sqrt{2\pi}}.$$

We have used the fact that

$$\frac{\mathbf{e}^{-(y-\beta_0-\beta_1 x)^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}$$

is a normal pdf and integrates to 1.

2. Dividing $f(x,y)$ by $f_X(x)$ gives

$$f_{Y|X}(x,y) = \frac{\mathbf{e}^{-(y-\beta_0-\beta_1 x)^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}.$$

The conditional distribution that we just obtained is a normal pdf with mean

$$\beta_0 + \beta_1 x$$

and variance σ^2.

The mean is an expected value (called the *conditional expectation*) and has the notation

$$E[Y|X = x] = \beta_0 + \beta_1 x.$$

The conditional expectation of Y, given $X = x$ is also referred to as the regression function, a function of x.

The variance is actually a *conditional variance*:

$$\textbf{Var}(Y|X = x) = \sigma^2.$$

This was the noise variance from before.

Independence

We have spoken earlier of cases where X_1 and X_2 are independent. In such cases, X_1 provided no predictive information about X_2; technically, this means that the conditional density function of X_2 given X_1 is identical to the marginal density function of X_2:

$$f_{X_2|X_1}(x_1, x_2) = f_{X_2}(x_2).$$

Multiplying both sides of this by $f_{X_1}(x_1)$ gives the joint density function on the left and the product of the marginal density function on the right. Random variables are independent if their joint distribution can be factored in this way. That is, X_1 and X_2 are independent, if their joint density is

$$f(x_1, x_2) = f_{X_1}(x_1) f_{X_2}(x_2) \tag{1}$$

where the two functions are the respective marginal densities of X_1 and X_2.

Graphical Views of Independence and Dependence

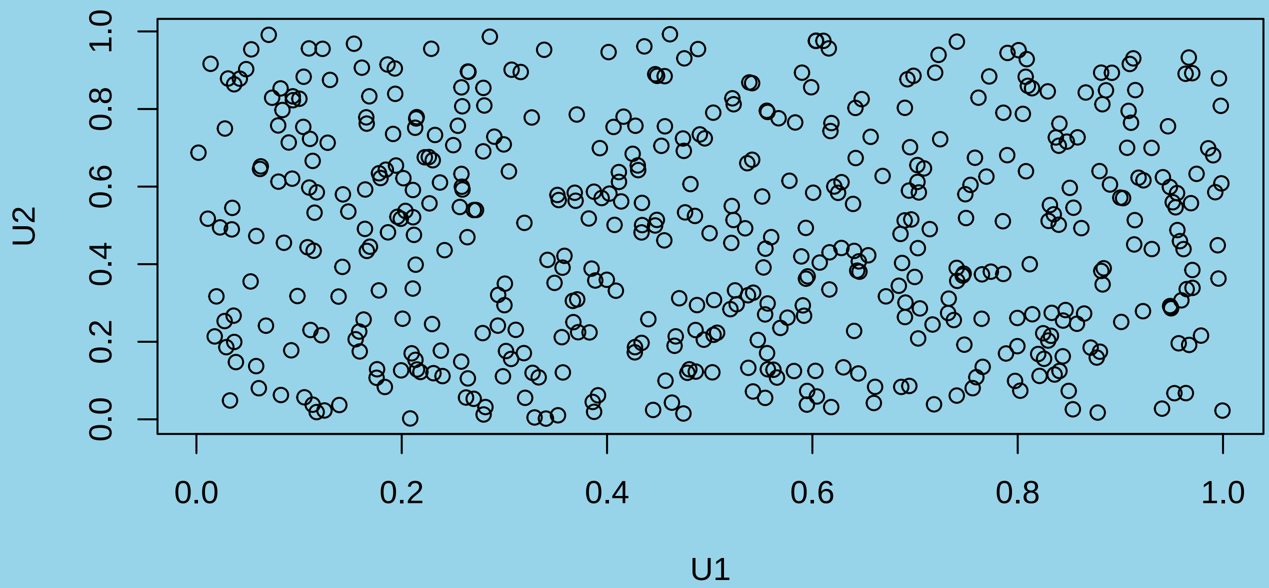
We can get an intuitive feel for (1) by graphing some simulated random variables, both independent and for some forms of dependence.

First, let's consider two independent uniform random variables which take values in the interval $[0, 1]$.

The next figure displays a scatterplot of 500 values taken from the distributions of U_2 and U_1, where are both uniformly distributed.

Graphical Views of Independence and Dependence

```
U1 <- runif(500)
U2 <- runif(500)
plot(U2 ~ U1)
```



There is no structure to the patterns that might be discerned from this picture. This is the clearest possible illustration of variables which have no relation.

Graphical Views of Independence and Dependence

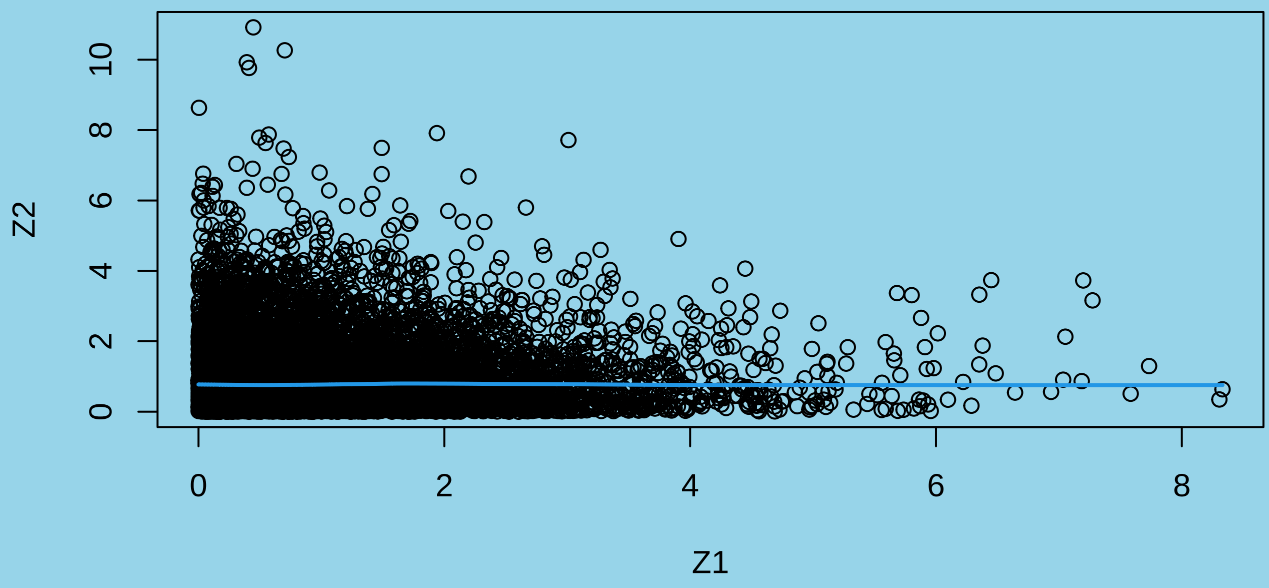
Independence manifests itself in other ways.

Next, we consider exponential random variables, Z_1 and Z_2.

In both cases, we will sample 10000 points from their respective distributions and look at a scatterplot of the corresponding pairs of data points.

```
Z1 <- rexp(10000);  Z2 <- rexp(10000)
plot(Z2 ~ Z1);  lines(lowess(Z1, Z2), col=4, lwd=2)
```

Graphical Views of Independence and Dependence



What you should observe in the figure is that it is not possible to predict the value of Z_2 from knowledge of Z_1.

This is a random collection of points, even though it might appear that there is a pattern (points are bunched up towards the lower left corner of the plot).

What characterizes independence is that neither variable gives predictive information about the other variable.

A Case of Dependence

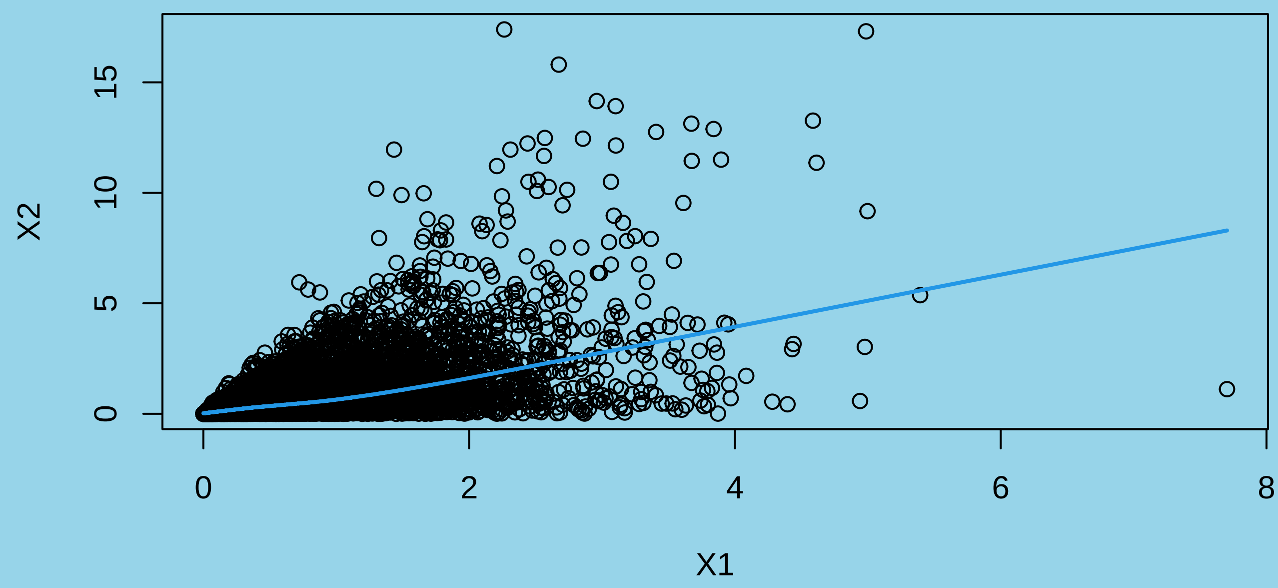
Let's change the story a bit now.

Suppose X_1 and X_2 are related. In particular, suppose X_1 is exponentially distributed with rate $\lambda = 1.5$ and X_2 is exponential with rate $1/X_1$.

Next, we will apply our simulation-based graphical analysis.

Graphical Views of Independence and Dependence

```r
X1 <- rexp(10000, rate = 1.5)
X2 <- rexp(10000, rate = 1/X1)
plot(X2 ~ X1)
lines(lowess(X1, X2), col=4, lwd=2)
```



Expected values are also calculated using double integrals:

$$E[g(X_1, X_2)] = \int \int g(x_1, x_2) f(x_1, x_2) dx_1 dx_2$$

where $g()$ is a continuous function of 2 variables, and the integrals are assumed to be taken over the support of the function $f()$ (i.e. wherever it is strictly positive).

For the uniform example above,

$$E[X_1 X_2] = \int_0^1 \int_0^1 x_1 x_2 dx_1 dx_2$$

$$= 0.25.$$

Expectation and Covariance - Another Example

Suppose

$$f(x_1, x_2) = \frac{\lambda}{x_1} e^{-\lambda x_1 - x_2/x_1}, \qquad x_1, x_2 \geq 0$$

and 0, otherwise.

$$E[X_1] = \lambda \int_0^\infty \int_0^\infty e^{-\lambda x_1 - x_2/x_1} dx_1 dx_2$$

$$= \lambda \int_0^\infty x_1 e^{-\lambda x_1} dx_1 = \frac{1}{\lambda}.$$

$$E[X_2] = \lambda \int_0^\infty \int_0^\infty \frac{x_2}{x_1} e^{-\lambda x_1 - x_2/x_1} dx_1 dx_2 = \frac{1}{\lambda}.$$

The *covariance* of two random variables gives a measure of their association, and can be calculated using

$$\text{Cov}(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2].$$

A nonzero value implies that the distributions of values of the pair of random variables will have (at least a vague) linear relationship.

If the covariance is positive, high values of X_1 will tend to be associated with high values of X_2 and low values of X_1 will be associated with low values of X_2.

When the covariance is negative, the opposite holds: high values of X_1 tend to occur with low values of X_2, and so on.

In the uniform example considered earlier, we see that Cov$(X_1, X_2) = 0$ which means that the two random variables do not have either a positive or negative linear association.

Dependent Exponential Random Variables

For another example, consider the random variables with the following joint probability density function

$$f(x_1, x_2) = \frac{\lambda}{x_1} e^{-\lambda x_1 - x_2/x_1}, \qquad x_1, x_2 \geq 0$$

and 0, otherwise. We can see that X_1 and X_2 are positively associated as follows.

$$E[X_1 X_2] = \lambda \int_0^\infty \int_0^\infty \frac{x_1 x_2}{x_1} e^{-\lambda x_1 - x_2/x_1} dx_1 dx_2 = \frac{2}{\lambda^2}.$$

To compute this integral, it is necessary to use integration-by-parts several times.

Thus,

$$\textbf{Cov}(X_1, X_2) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

This value is positive which means that X_1 and X_2 are positively related.

Correlation

The magnitude of the covariance, when nonzero, depends on the scale of the variables.

For example, if the covariance between X_1 and X_2 is 1.0, it can be shown that the covariance between $10X_1$ and $10X_2$ is 100.

By dividing through by the standard deviations of both variables, we can remove this dependence on scale. The resulting measure is the *correlation*:

$$\textbf{Corr}(X_1, X_2) = \frac{\textbf{Cov}(X_1, X_2)}{\sqrt{V(X_1)V(X_2)}}.$$

When the covariance is 0, the correlation will obviously be 0, so this is the case for the two error proportion measurements discussed earlier: we say these measurements are *uncorrelated*.

Correlation

Positive covariances lead to correlations which are positive but no larger than 1. A scatterplot of positively correlated variables should show points that scatter about a line with a positive slope.

Negative covariances lead to correlations which are negative but no less than -1. A scatterplot of negatively correlated variables should show points that scatter about a line with a negative slope.

Calculation of covariance and correlation for a sample

For a sample $\{(x_{11}, x_{21}), (x_{12}, x_{22}), \ldots, (x_{1n}, x_{2n})\}$, the *sample covariance* is given by

$$c = \frac{1}{n-1} \sum_{j=1}^{n} (x_{1j} - \bar{x}_1)(x_{2j} - \bar{x}_2).$$

The *sample correlation* is given by

$$r = \frac{c}{s_1 s_2}$$

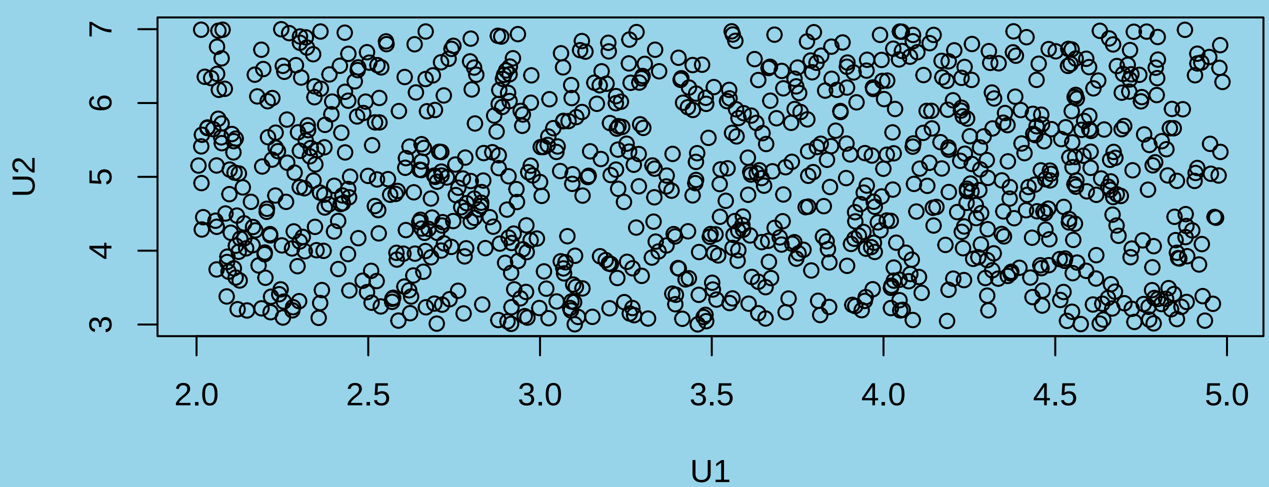
where s_1 and s_2 are the sample standard deviations of the samples of x_1's and x_2's respectively.

Correlation - Example

Independent Uniforms:

```
U1 <- runif(1000, 2, 5)
U2 <- runif(1000, 3, 7)
cor(U1, U2); plot(U2 ~ U1)

## [1] 0.006099031
```

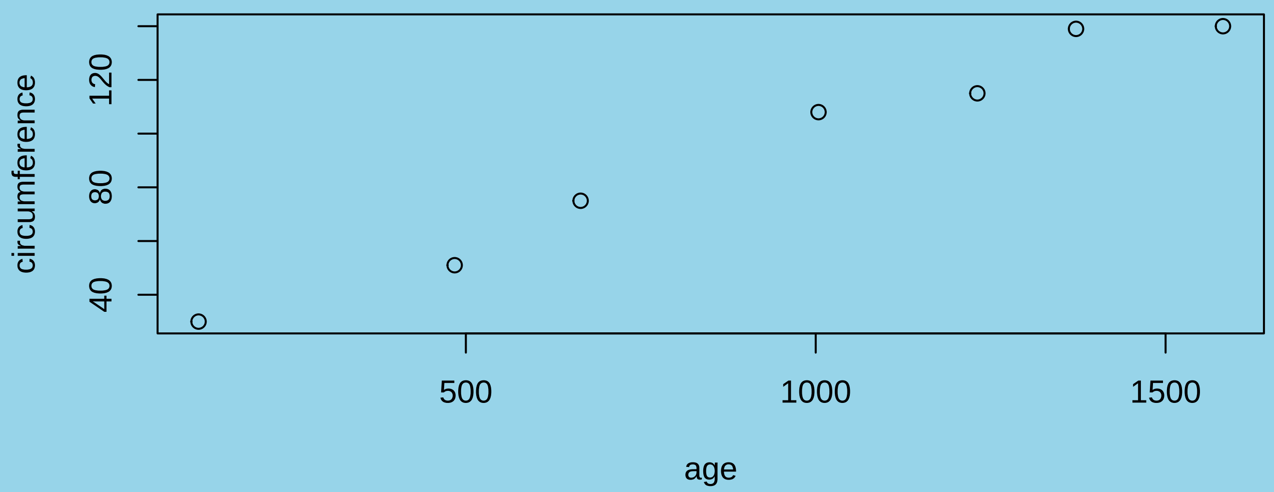


The correlation is small, close to 0, and the scatterplot shows no pattern.

Correlation - Example

Orange tree circumference versus age (`Orange` data frame)

```
Orange3 <- subset(Orange, Tree == 3) # data on Tree No. 3
plot(circumference ~ age, data = Orange3)
```




```
with(Orange3, cor(circumference, age))

## [1] 0.9881766
```

The correlation is large and positive, and the points scatter about a line with positive slope.

Correlation versus Dependence

What is the difference between correlation and dependence?

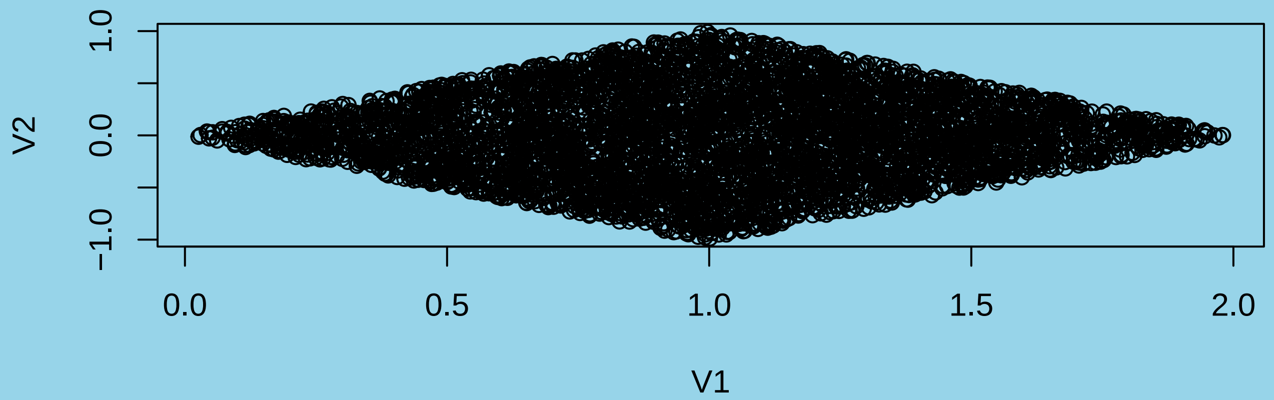
Correlation ... is a measure of *linear* dependence.

2 random variables may be dependent, but not correlated.

Example

```r
U1 <- runif(5000); U2 <- runif(5000)
V1 <- U1 + U2;  V2 <- U1 - U2
cor(V1, V2); plot(V2 ~ V1)

## [1] 0.02328041
```



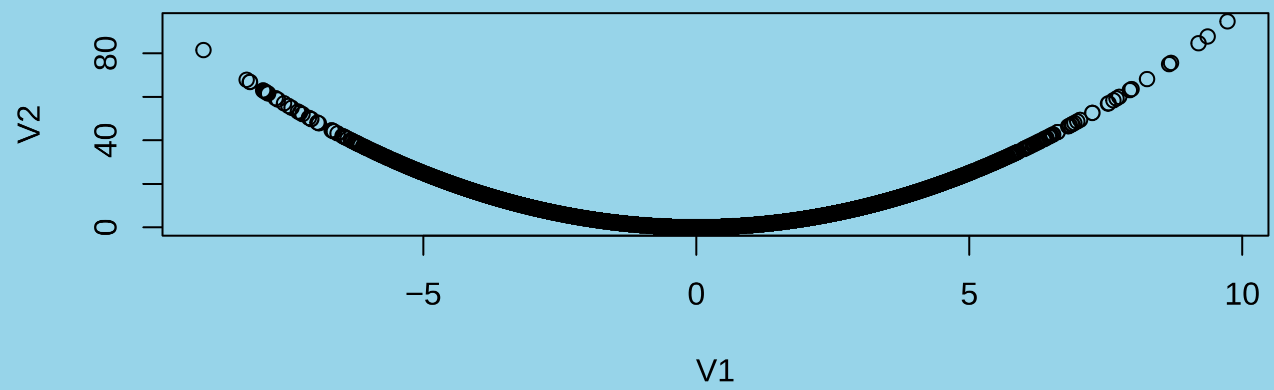
Are V_1 and V_2 dependent?

Are V_1 and V_2 correlated?

Example 2

```
U1 <- rexp(50000); U2 <- rexp(50000)
V1 <- U1 - U2;  V2 <- V1^2
cor(V1, V2);  plot(V2 ~ V1)

## [1] 0.01284874
```



Are V_1 and V_2 dependent?

Are V_1 and V_2 correlated?

What to Take Away from this Lecture

Modelling several variables at the same time can be complicated.

The assumption of independence greatly simplifies the problem by allowing consideration of each variable separately – independently of all of the other variables.

If the independence assumption is used, it must be checked:

- **graphically - familiarize yourself with what a random pattern looks like; if a nonrandom pattern appears on a scatterplot of two or three random variables, they cannot be independent.**
- **numerically - the correlation coefficient can indicate *linear* dependence**

What to Take Away from this Lecture

Dependence can also be very useful: if a random variable is dependent on other random variables, the other variables can be used for prediction.

The conditional probability density function is essentially a predictive function.

Regression modelling is based on the expected value of a random variable, using the conditional probability density function.